

QUESTION 1 [17 marks]

A system consists of five (5) processes (P0 to P4) and three (3) resource types (A, B, C), where each resource type has multiple instances. The total number of instances for each resource type is as follows: 12 instances of resource A, 7 instances of resource B, and 8 instances of resource C. The allocation matrix and the maximum requirement matrix for the processes are provided in **Table 1** below:

Table 1

Process	Allocation			Maximum Requirement		
	A	B	C	A	B	C
P0	2	1	1	6	4	3
P1	3	0	2	5	2	4
P2	1	1	1	4	2	2
P3	0	2	1	2	3	4
P4	2	0	2	6	1	5

Table 1

Using the information above and applying the Banker's Algorithm, answer the following questions:

- Calculate the current available resources for each resource type (A, B, C). Clearly show your calculations.[2 marks]
- Calculate the Need matrix using the given allocation and maximum requirement matrices. Clearly show your calculations for each process.[3 marks]
- Determine whether the system is currently in a safe state using Bank's Algorithm. If the system is in a safe state, provide the safe sequence. [7 marks]
- Assume that process P1 requests additional resources (2, 2, 1). Determine whether the system can allocate these resources to P1. Provide clear computations and reasoning in your answer.[5 marks]

QUESTION 2 [15 marks]

A. A system uses a contiguous memory allocation mechanism for managing its physical memory. The total physical memory space is 1000 KB, and it is divided among four processes (P1, P2, P3, and P4). The base address and limit for each process are given in **Table 2**.

Table 2: Memory Allocation for Processes

Process	Base	Limit
P1	0	200
P2	200	300
P3	500	150
P4	650	250

Table 2

Determine the physical memory address for the logical addresses: P2:250 and P4:300. Show all your workings and justify whether the addresses are valid based on the given memory limits for each process. [4 marks]

- (b) A system has a physical memory of 1 MB and is divided into five partitions. The sizes of the partitions are listed in **Table 3**.

Table 3: Memory Partitions

Partition	1	2	3	4	5
Size (KB)	250	150	300	200	100

Consider four jobs (J1, J2, J3, J4) with different memory requirements, as shown in **Table 4**. Allocate these jobs to memory partitions in order using the first-fit, best-fit, and worst-fit allocation algorithms. Complete **Table 5** by writing the memory partition numbers for each job under each algorithm. [4 marks]

Table 4: Job Sizes

Job	Size (KB)
J1	200
J2	120
J3	260
J4	100

Table 5: Memory Allocation Results

(Copy the table to answer booklet)

Job	First-fit	Best-fit	Worst-fit
J1			
J2			
J3			
J4			

- (c) Consider a paging system for a logical address space of 2^6 bytes with a page size of 2^3 bytes and a physical memory of 256 bytes. Answer the following questions. Show your working.

- What is the size of each frame in the physical memory? [1 mark]
- How many frames are available in the physical memory? [2 marks]
- How many entries are required in the page table? [2 marks]
- If the frame number for data is 6 and the offset is 4, calculate the physical address. [2 marks]

QUESTION 3 [18 marks]

Given the following reference string:

5	2	0	1	2	0	5	3	6	3	0	1	2	0	6	0	3	1	2	3	4	1	2
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- (a) Identify the total number of page faults using First-In-First Out (FIFO) Page Replacement and 4-page frames. Show all your workings. [5 marks]
- (b) Identify the total number of page faults using Optimal Page Replacement (OPT) and 4-page frames. Show all your workings. [5 marks]
- (c) Identify the total number of page fault using Least Recently Used (LRU) page replacement and 4-page frames. Show all your workings. [5 marks]
- (d) Based on your answers to Questions 1(a), 1(b), and 1(c), identify the least effective replacement technique. Explain why it is the worst choice and describe two consequences of having frequent page faults. [3 marks]

QUESTION 4 [15 marks]

- (a) A file system uses indexed allocation with a linked scheme to manage disk space. The following block addresses represent the content of a file:

4, 15, 8, 12, 20, 6, 14, 2, 9, 11, 18, 5, 13, 3, 7, 10, 17, 1, 16, 19

Assume the following:

- Each index block can hold 5 block addresses.
- When one index block is full, the last address will store the pointer to the next index block.

List down all index blocks and draw the structure of the blocks used to store the file using an indexed allocation with a linked scheme. Label each index block and clearly show its block addresses. [5 marks]

- (b) A file system uses Contiguous Allocation to store files on the disk. Consider a file F1 of size 15 blocks, starting at block 12. Another file F2 of size 10 blocks follows immediately after F1.
- i. Draw the allocation of blocks on the disk for both files. [3 marks]
 - ii. Calculate the external fragmentation if block sizes are 4KB, and a free space of 5 consecutive blocks exists after file F2. [2 marks]
- (c) The following bit map is used for free space management in a hypothetical Operating System. The system uses a bit value of 1 to represent a free block and a bit value of 0 to represent a used block. Each word in the bit map contains 8 bits.

The shaded block indicates the **6th word** in the bit map in **Figure 2**.

Block No	0	1	2	3	4	5	6	7	8	...											
Bit Map	0	0	0	0	0	0	0	0	0	...	0	0	0	0	1	1	0	0	0	0	...
Offset	0	1	2	3	4	5	6	7	0	...	0	1	2	3	4	5	6	7	0	1	...

Figure 2

Find the address of the **first free block** in the bit map. Show all your calculations step by step.

[5 marks]

QUESTION 5 [15 MARKS]

A disk drive with 200 cylinders numbered from 0 to 199 has a queue of requests for disk arm movement at the following cylinder positions:

[34, 120, 12, 180, 75, 20, 150, 60]

The initial position of the disk arm is at cylinder **90**.

For the following disk scheduling algorithms, illustrate the movement with a graph and calculate the **total distance** the disk arm travels to satisfy all the requests. Show the order of access and step-by-step calculations for each algorithm:

- (a) Shortest Seek Time First (SSTF) [5 marks]
- (b) C-SCAN (outward direction: inner → outer) [5 marks]
- (c) C-LOOK (outward direction: inner → outer) [5 marks]

- End of question -